Quantification: I534_z19c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	18
interior lacuna count	18
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2991.17	2662	1765.29	px^2
major_axis_length_px	lacuna shape	92.6792	85.7246	38.3527	px
minor_axis_length_px	lacuna shape	47.3786	46.3695	13.7194	px
aspect_ratio	lacuna shape	1.9825	2.121	0.5927	unitless
eccentricity	lacuna shape	0.7911	0.8819	0.1778	unitless
solidity	lacuna shape	0.737	0.7458	0.113	unitless
roots_count	headline	12.1667	12.5	4.7558	count
ring_length_r30_px	headline	313.167	304.5	97.0708	px
ring_length_r60_px	reported	622.111	616	164.098	px
owned_length_px	ownership-dependent	764.207	555.109	593.311	px
edge_count	ownership-dependent	36.1667	24	27.9332	count
mean_edge_length_px	ownership-dependent	21.1606	20.7825	3.2283	px
perimeter_px	normalised	320.019	286.099	133.596	px
ring_area_r30_px2	normalised	10753.7	10157.5	3274.58	px^2
ring_area_r60_px2	normalised	22675	21967.5	4299.92	px^2
in_frame_fraction_r30	normalised	0.9974	1	0.011	unitless
in_frame_fraction_r60	normalised	0.985	1	0.0438	unitless
ring_density_r30	normalised	0.0302252	0.0297333	0.01063	px^-1
ring_density_r60	normalised	0.0281354	0.026588	0.0088417	px^-1
roots_per_100px_perimeter	normalised	4.0662	3.8968	1.679	per 100 px
ring_attached_length_r30_px	network v2	281.889	271.5	92.9047	px
ring_attached_length_r60_px	network v2	446.056	454	171.458	px
ring_length_w_r30_px	network v2	355.873	345.25	110.209	px
ring_length_w_r60_px	network v2	706.342	705.926	186.628	px
sholl_crossings_r10	network v2	7.1667	7.5	2.1213	count
sholl_crossings_r20	network v2	8.0556	8.5	3.2262	count
sholl_crossings_r30	network v2	7.8889	8	2.6097	count
loop_count	loops, report only	1.6111	0	3.0513	count
loop_edge_share	loops, report only	0.0914	0	0.1208	share
ring_width_mean_r30_px	width	7.4987	7.4813	0.5432	px

Per field
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measure	value	units
analysed_area_px2	845861	px^2
total_skeleton_length_px	21316	px
canalicular_length_density_per_px	0.0252004	px^-1
junction_count	412	count
junction_density_per_px2	0.00048708	px^-2
lacuna_count	18	count

lacunae_per_px2	2.128e-05	px^-2
skeleton_component_count	308	count
mean_component_length_px	69.2078	px
median_component_length_px	27	px
field_length_density_w_per_px	0.0285792	px^-1
field_density_without_flagged_per_px	0.0213723	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	418	px
unattached_component_count	24	count
canalicular_length_density_attached_per_px	0.0252004	px^-1
field_length_density_with_unattached_per_px	0.0256945	px^-1
excluded_area_px2	148874	px^2
analysed_field_area_px2	899702	px^2
loop_count	39	count
loop_edge_share	0.1195	share
edge_count	1205	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.0316	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	21797	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1823	56.4889	49.8735	1.1326	0.4696	0.8271	-1.1579	125.003	757.529	False
2	1637	57.8693	48.4179	1.1952	0.5477	0.7276	-0.0784	295.51	896.101	False
3	1291	47.4215	38.1837	1.2419	0.593	0.8741	0.4227	324.111	336.189	False
4	6026	129.945	61.3358	2.1186	0.8816	0.8485	-1.1185	377.349	511.866	False
5	1342	66.6735	36.4005	1.8317	0.8378	0.6286	1.399	396.13	787.283	False
6	3334	115.437	53.1764	2.1708	0.8876	0.6441	-0.8143	448.481	612.295	False
7	4876	84.2608	78.5547	1.0726	0.3617	0.7525	0.8631	503.31	62.7231	False
8	3737	84.848	59.7745	1.4195	0.7097	0.8338	-0.4377	493.364	441.851	False
9	820	51.4532	24.5895	2.0925	0.8784	0.7614	1.0688	505.329	211.1	False
10	3553	104.824	44.321	2.3651	0.9062	0.8721	0.2384	581.822	856.943	False
11	6397	200.128	63.1285	3.1702	0.9489	0.4898	1.452	595.701	654.416	False
12	2632	101.535	38.4077	2.6436	0.9257	0.6745	-1.3449	587.391	245.703	False
13	5166	140.866	61.0892	2.3059	0.9011	0.6428	0.013	686.162	69.5337	False
14	1712	87.747	36.8261	2.3827	0.9077	0.6786	-0.5403	724.265	671.19	False
15	914	54.9878	33.7201	1.6307	0.7899	0.5745	-0.9811	788.83	430.015	False
16	2692	86.6013	40.7838	2.1234	0.8822	0.8628	-1.1084	820.32	99.4658	False
17	1680	84.8432	32.3344	2.6239	0.9245	0.7391	-0.8417	827.977	860.805	False
18	4209	112.297	51.8973	2.1638	0.8868	0.8336	-0.6033	855.455	275.629	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	4	143	525	341.593	17	20.0937	7.2097
2	7	137	288	403.09	20	20.1545	6.609
3	9	269	718	2573.1	121	21.2653	6.7737
4	20	382	643	558.345	36	15.5096	7.0925
5	12	445	969	1745.76	86	20.2996	7.6746
6	13	287	421	327.078	19	17.2146	8.0044
7	16	421	656	883.465	37	23.8774	7.4938
8	15	237	506	307.764	17	18.1038	7.4688
9	13	385	605	755.097	48	15.7312	8.3586
10	15	328	887	1286.22	58	22.1762	8.1696
11	21	480	779	551.872	24	22.9947	7.8626
12	17	368	607	475.86	21	22.66	7.4204
13	14	404	623	335.735	17	19.7491	7.9182
14	8	322	608	915.293	35	26.1512	7.1883
15	12	249	609	952.377	38	25.0625	7.5833
16	8	236	400	298.392	15	19.8928	6.9221
17	6	271	659	462.676	18	25.7042	8.3868
18	9	273	695	582.014	24	24.2506	6.8395

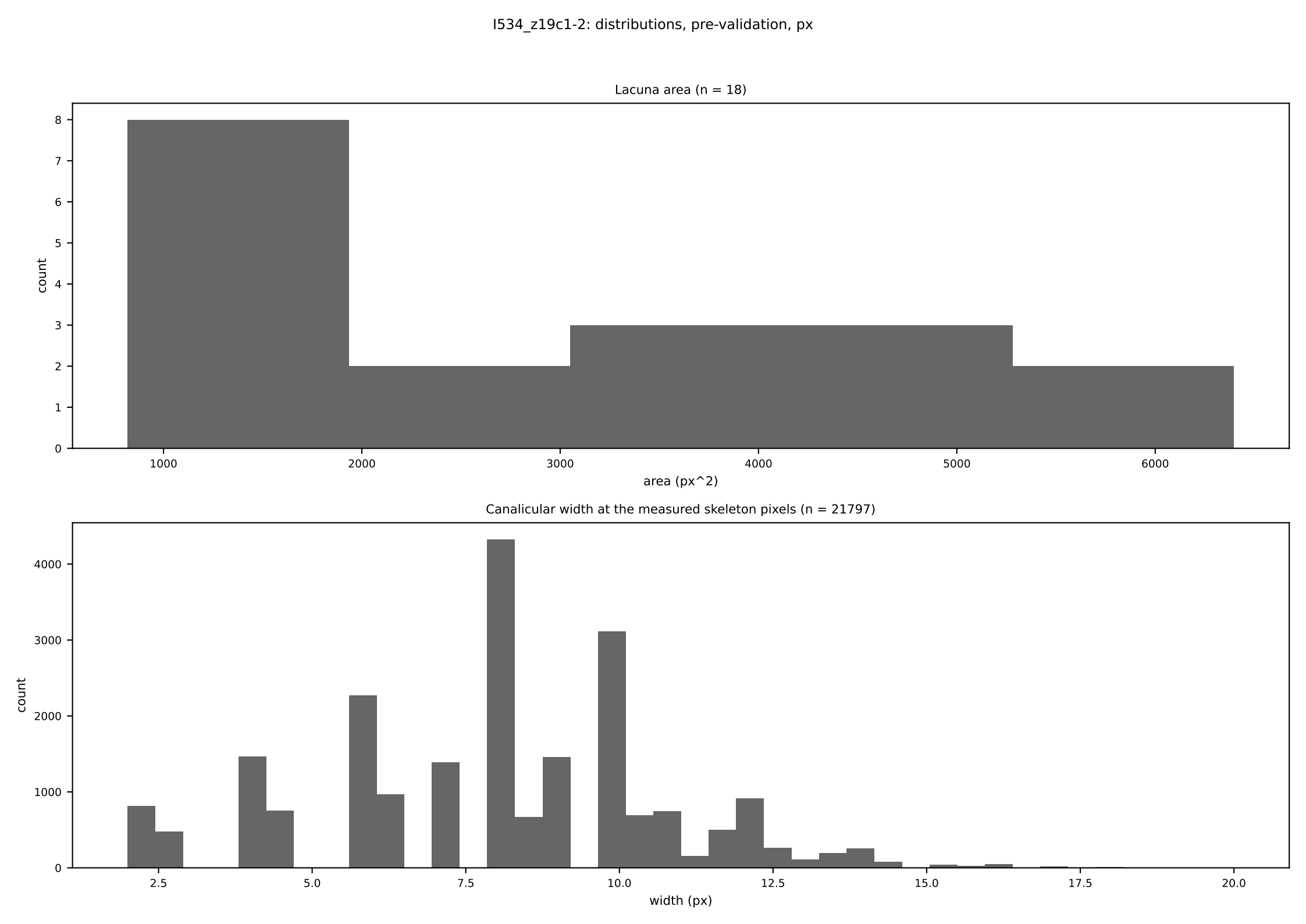
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	272.936	8396	22205	1	1	0.0170319	0.0236433	1.4655
2	283.463	8668	21456	1	1	0.0158053	0.0134228	2.4695
3	164.51	7220	19561	1	1	0.0372576	0.0367057	5.4708
4	382.854	12834	23401	1	1	0.0297647	0.0274775	5.2239
5	262.871	8654	20100	1	1	0.0514213	0.048209	4.565
6	368.475	12793	22015	1	1	0.0224341	0.0191233	3.5281
7	375.103	12127	21219	0.9532	0.8444	0.0347159	0.0309157	4.2655
8	300.049	10385	21896	1	1	0.0228214	0.0231092	4.9992
9	155.717	6688	13329	1	1	0.0575658	0.0453897	8.3485
10	288.735	10625	24527	1	1	0.0308706	0.0361642	5.1951
11	725.092	20326	35311	1	1	0.0236151	0.0220611	2.8962
12	316.777	10912	23620	1	1	0.0337243	0.0256986	5.3666
13	533.263	15650	25399	1	0.8882	0.0258147	0.0245285	2.6253
14	263.664	9930	21920	1	1	0.032427	0.0277372	3.0342
15	215.723	8071	20618	1	1	0.0308512	0.0295373	5.5627
16	247.279	9530	20421	1	0.9977	0.0247639	0.0195877	3.2352
17	251.22	9124	23689	1	1	0.0297019	0.0278188	2.3883
18	352.617	11633	27463	1	1	0.0234677	0.0253068	2.5523

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	126	205	163.853	608.299	4	7	7
2	108	149	152.024	317.421	4	1	4
3	232	469	305.35	816.796	4	9	9
4	326	486	428.433	725.056	8	12	10
5	415	810	504.404	1097.26	10	13	12
6	256	280	325.492	476.274	6	5	4
7	409	615	486.931	745.612	10	11	10
8	222	282	265.794	568.973	6	5	4
9	341	541	434.635	679.487	8	12	7
10	304	679	365.007	996.666	7	9	13
11	454	595	540.073	889.507	11	5	9
12	317	405	418.647	691.512	8	10	7

13	343	430	460.546	711.583	9	11	10
14	287	439	376.676	700.269	8	9	8
15	249	486	288.836	691.127	5	4	5
16	199	250	270.007	457.517	6	8	7
17	248	518	298.309	729.6	8	7	8
18	238	390	320.706	811.193	7	7	8


I534_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 18)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 21797)
count
width (px)